ELL-ADIC COMPLETION PRESERVES EXACTNESS OF A SHORT SEQUENCE OF ABELIAN GROUPS UNDER A FINITENESS HYPOTHESIS

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Allowed standard prerequisites (editorial).

- Right exactness of tensoring with Z/ell^m and the surjectivity of the induced map on completions. Used inside the counted proof at lines 1093-1102. Produces exactness of A/ell^m -> B/ell^m -> E/ell^m -> 0 and the Mittag-Leffler style surjectivity of the transition maps.
- Snake lemma and induction on m for the finiteness of the ell-power torsion. Used at lines 1104-1125. Shows that _{ell^m}E is finite for all m and produces the exact sequence whose projective limit is the desired one.

Lemma 1.1. *Let $A \to B \to E \to 0$ be an exact sequence of abelian groups.*

(a) *The induced map $B^{(\ell)} \to E^{(\ell)}$ is surjective.*

(b) *If ${}_\ell E$ is finite, then the induced sequence*

$$A^{(\ell)} \to B^{(\ell)} \to E^{(\ell)} \to 0$$

is exact. Likewise if A/ℓ is finite.

Proof. (a) Since the functor $. \otimes_{\mathbf{Z}} \mathbf{Z}/\ell^m$ is right exact, the sequence

$$A/\ell^m \to B/\ell^m \to E/\ell^m \to 0$$

is exact. Therefore, the projective system $(\ker[B/\ell^m \to E/\ell^m])_{m\geq 1}$ has surjective transition maps, which implies that the map $\varprojlim_m (B/\ell^m) \to \varprojlim_m (E/\ell^m)$ remains surjective.

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(b) Assume that ${}_\ell E$ is finite. Then (by induction on m) we also have that ${}_{\ell^m} E$ is finite for every positive integer m thanks to the exact sequence

$$ {}_\ell E \to {}_{\ell^{m+1}} E \xrightarrow{\cdot \ell} {}_{\ell^m} E. $$

Let $I \subset B$ be the image of A by the map $A \to B$. By (a), the map $A^{(\ell)} \to I^{(\ell)}$ is surjective, so it is sufficient to prove that the sequence

$$ I^{(\ell)} \to B^{(\ell)} \to E^{(\ell)} $$

is exact. By the snake lemma, we have an exact sequence

$$ {}_{\ell^m} E \to I/\ell^m \to B/\ell^m \to E/\ell^m. $$

Taking projective limit over m yields the required exact sequence because the kernel of the map $I/\ell^m \to B/\ell^m$ is finite (it is a quotient of ${}_{\ell^m} E$). Similarly, if A/ℓ is finite, then A/ℓ^m (hence also I/ℓ^m) is finite for every positive m and the same argument works. □

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